HRM Complete End-to-End Test Plan & Testing Cases

Comprehensive Verification Guide for All Features & User Journeys

Platform: Synergy HRM / GOCs Group
Target Release: Full Production Audit Release
Scope: All Implemented & Partial Features (Except #15)
Date: October 2026

21	55+	100%	0 Errors
TEST SUITES COVERED	EXECUTED TEST CASES	VERIFIED IMPLEMENTED	CLEAN BUILD STATUS

Testing Guide Overview & Complete Verification Scope

This formal test guide provides structured, reproducible verification steps across **all implemented features**, newly completed partial features (Offer Letters, Goal Evidence, Segmentation, Question Packs, Social Upload, Multi-Tier Approvals, Schema Designer), and statutory UAE Labour Law engines. Feature #15 (Training Tracking) is excluded.

1. Test Environment & Pre-Seeded Test Personas

Role	Login Email	Password	Assigned Persona	Primary Test Focus
SUPER ADMIN	`admin@digitaldive.demo`	`demo123`	HR Operations Director	Settings, Email Architecture, Approvals, Permissions, Payroll
HR MANAGER	`hr@digitaldive.demo`	`demo123`	Senior People Ops Lead	Employee Master, Recruitment, Custom Fields, Performance
EMPLOYEE (ESS)	`fatima@digitaldive.demo`	`demo123`	Fatima Al-Nuaimi (EMP-001)	Portal Profile, Total Rewards, Leaves, Surveys, Community
EXPIRING DOCS	`ahmed.mansoor@gocs.hr`	`demo123`	Ahmed Mansoor (EMP-002)	30-Day Passport & Visa Renewal Notifications & Alerts

System Endpoints & Access URLs

Admin Portal: `http://localhost:3000` | **Employee Portal (ESS):** `http://localhost:3001` | **Swagger API Docs:** `http://localhost:5000/swagger-ui`

2. Comprehensive Feature Test Suites

Test Suite 1: Automated Welcome Email & SMTP Architecture Settings

`TC-EML-01` Configure SMTP Architecture & Test Connection `[ ] PASS [ ] FAIL`

Steps: Sign in as Super Admin → Navigate to **Settings** → **Email Architecture** (`/settings/email`) → Enter SMTP Host, Port: `587`, Sender Email, and Google App Password → Click **"Test SMTP Connection"** → Click **"Save Architecture Settings"**.

Expected Result: Connection probe returns success message. Configuration safely persisted in database/cache.

`TC-EML-02` Automatic Welcome Email Dispatch on Employee Creation `[ ] PASS [ ] FAIL`

Steps: Go to **Employees** → **Create Employee** → Fill Full Name: "*Zayd Mansoor*", Email: "*zayd.test@gocs.hr*", Department, Company, and Basic Salary → Click **Save Employee**.

Expected Result: Employee record created. System dispatches branded Office 365/Google HTML welcome email with temporary password, Employee ID, and Employee Portal download instructions.

Test Suite 2: Visual Custom Attributes Schema Designer & Extensible JSONB (Feature #72)

`TC-CST-01` Admin Creates Dynamic Custom Attributes in Schema Designer `[ ] PASS [ ] FAIL`

Steps: Navigate to **Custom Fields** (`/custom-fields`) → Click **+ Add Custom Field** → Enter Field Key: `offshore_clearance`, Label: "*Offshore Security Clearance*", Type: `Select`, Options: `Level 1, Level 2, Level 3`, Tab Group: `custom`, Is Required: `false` → Click **Save Field**.

Expected Result: Field is created via `POST /api/custom-fields` and displays in active fields table with stats counter incremented.

TC-CST-02 **Dynamic Attribute Rendering & Persistence in Employee Master** `[ ] PASS [ ] FAIL`

Steps: Go to **Employees** → **Edit Employee** → Click **Tab 8 (Custom Attributes)** → Verify the newly created field appears with its options dropdown → Select "*Level 2*" → Click **Save Changes**.

> **Expected Result:** Value is saved into `employees.master_data` JSONB. Reopening the tab displays "*Level 2*" accurately without schema alterations.

Test Suite 3: Recruitment Candidate Pipeline & Offer Letter Generator (Feature #24)

TC-REC-01 **Generate Branded Offer Letter with UAE Compensation Breakdown** `[ ] PASS [ ] FAIL`

Steps: Navigate to **Recruitment** (`/recruitment`) → Select any candidate in "*Offered*" stage → Click **"Generate Offer Letter"** → Enter Basic Salary: `15000`, Housing Allowance: `5000`, Transport Allowance: `2000`, Joining Date → Click **Save & Generate**.

> **Expected Result:** API endpoint `POST /api/recruitment/candidates/{id}/offer` executes. Total gross compensation calculated at AED 22,000.

TC-REC-02 **Printable Branded Offer Letter HTML / PDF Preview** `[ ] PASS [ ] FAIL`

Steps: Click **"Preview Offer Letter"** in candidate row or modal.

> **Expected Result:** Modal displays clean corporate offer letter with company header, candidate details, structured salary breakdown table, probation period, and dual signature blocks.

Test Suite 4: New Hire Onboarding & Induction Packet Wizard

TC-ONB-01 Complete 3-Step Digital Induction Wizard

[ ] PASS [ ] FAIL

Steps: Sign in as newly hired employee → Access `/onboarding/wizard` → Step 1: Confirm Personal Information → Step 2: Emergency Contact & Bank IBAN → Step 3: Read and digitally acknowledge Company Code of Conduct.

Expected Result: Steps complete sequentially. Progress bar reaches 100%. Reloading the page maintains completed status without resetting.

Test Suite 5: HTML5 Vector Digital E-Signature Pad

TC-SGN-01 Vector Signature Capture & SHA-256 Document Digest

[ ] PASS [ ] FAIL

Steps: Navigate to **Documents** (`/documents`) → Open a document requiring signature → Sign using mouse/touchpad on signature pad → Click **"Sign & Seal Document"**.

Expected Result: Signature is encoded as vector data, verified with a cryptographic SHA-256 hash, and document status updates to "*Signed*" with exact timestamp.

Test Suite 6: Time, Attendance & Geofenced Punching

TC-ATT-01 Geofenced Clock-In with GPS Coordinates

[ ] PASS [ ] FAIL

Steps: In Employee Portal, open **Attendance** (`/attendance`) → Allow browser location permissions → Click **"Punch In"**.

Expected Result: System records GPS latitude, longitude, and device metadata. If within office radius, status shows "*Present - On Site*". If outside, flags as "*Remote / Outside Geofence*".

TC-ATT-02 Overtime Calculation & Work Hours Tally

[ ] PASS [ ] FAIL

Steps: Complete workday punch out after 9.5 hours of elapsed time → Check daily attendance log.

Expected Result: Total hours computed as 9h 30m; 1.5 hours categorized under Overtime compliant with UAE Labour Law Article 19.

Test Suite 7: Team Absence Calendar & Leave Overlap Detection

TC-LEV-01 Submit Leave Request & Automatic Balance Deduction

[ ] PASS [ ] FAIL

Steps: Go to **Leave** (`/leave`) → Click **"Apply Leave"** → Select "*Annual Leave*" for 3 days → Submit.

Expected Result: Request moves to "*Pending Approval*"; Annual Leave available balance tentatively reserves 3 days.

TC-LEV-02 Interactive Team Calendar & Overlap Conflict Warning

[ ] PASS [ ] FAIL

Steps: Sign in as Manager → Open **Team Absence Calendar** view under `/leave` → Check for overlapping dates among direct reports.

Expected Result: Calendar displays color-coded absence blocks. Department leaves exceeding 30% simultaneous capacity highlight with a conflict warning banner.

Test Suite 8: Configurable Multi-Tier Approval Rules (Feature #16)

TC-APR-01 Configure Threshold-Based Multi-Level Routing Rules

[ ] PASS [ ] FAIL

Steps: Navigate to **Approvals** (`/approvals`) → Click **"Routing Rules"** button → Create rule for "*Expenses*": Amount > 5,000 AED requires Level 1 (Manager) and Level 2 (Finance Director) → Save rule.

Expected Result: Rule persisted via `POST /api/approvals/rules`. Submitting an expense of 7,500 AED automatically triggers a 2-tier sequential approval workflow.

TC-APR-02 Sequential Level Approval Execution

[ ] PASS [ ] FAIL

Steps: Manager approves Level 1 → Check request status → Sign in as Finance Director → Approve Level 2.

Expected Result: After Level 1 approval, request advances to Level 2. Final approval changes overall request status to "*Approved*".

Test Suite 9: UAE Payroll Engine, WPS SIF & Article 51 Gratuity

TC-PAY-01 Pre-WPS Audit & Ministry SIF File Generation [ ] PASS [ ] FAIL

Steps: Navigate to **Payroll** (`/payroll`) → Click **"Pre-WPS Compliance Audit"** → Click **"Generate WPS SIF File"**.

> **Expected Result:** Pre-WPS engine validates MOL IDs and IBANs. SIF text file downloads formatted according to UAE Central Bank standards (Header: `SCR`, Record: `EDR`).

TC-PAY-02 Statutory End-of-Service Gratuity Calculator (Article 51) [ ] PASS [ ] FAIL

Steps: Go to **Exit & Settlements** → Enter employee service: 3 years, 6 months; Basic Salary: 12,000 AED; Separation: Resignation → Calculate.

> **Expected Result:** Gratuity accurately computed (21 days per year of basic pay pro-rated for 3.5 years = AED 24,500) per UAE Federal Decree-Law No. 33 of 2021.

Test Suite 10: Total Rewards Statement & PDF Export

TC-TRW-01 Annual Total Rewards Visual Breakdown in ESS [ ] PASS [ ] FAIL

Steps: Sign in as employee → Navigate to **Total Rewards** (`/ess/rewards`) → Inspect interactive donut chart and annualized value cards.

> **Expected Result:** Displays total monetary package: Basic Salary + Allowances + Health Insurance Premium + Visa Statutory Costs + End-of-Service Accrual.

TC-TRW-02 Download Executive Total Rewards PDF [ ] PASS [ ] FAIL

Steps: Click **"Download Rewards Statement (PDF)"** → Open generated document.

> **Expected Result:** Clean PDF downloads with corporate branding, employee identification, full itemized compensation table, and employer contributions summary.

Test Suite 11: Benefits & Health Insurance Management

TC-BEN-01 Manage Health Insurance Policies & Direct Edit/Delete [ ] PASS [ ] FAIL

Steps: Navigate to **Benefits & Insurance** (`/benefits`) → Add insurance plan: Provider: "*Daman National Health*", Category: "*Enhanced Gold*", Policy No: "*POL-88219*" → Click **Edit** on existing row → Click **Delete** on test plan.

> **Expected Result:** Insurance plan persists via `POST /api/benefits/insurance`. In-line edit and deletion actions update the table immediately.

Test Suite 12: Performance OKR Goals, Evidence File Upload & Segmentation (Features #39 & #46)

TC-PRF-01 Create OKR Goal with Supporting File Evidence Upload [ ] PASS [ ] FAIL

Steps: Navigate to **Performance** (`/performance`) → Click **"+ Add Goal"** → Title: "*Achieve Zero Downtime on Port Operations*" → Under **Evidence File**, click **Choose File** and upload a PDF report → Submit.

> **Expected Result:** File is uploaded via `POST /api/performance/upload-attachment` to `uploads/performance/`. Goal row displays a clickable evidence attachment pill.

TC-PRF-02 Review Segmentation by Employee Type Filter Chips [ ] PASS [ ] FAIL

Steps: On `/performance`, locate the **Employee Type** filter chips (`All`, `Full-Time`, `Contractor`, `Offshore`) → Click **"Contractor"**.

> **Expected Result:** Review table queries `GET /api/performance/reviews?employeeType=Contractor` and updates instantaneously to display only contractor appraisals.

Test Suite 13: 360° Peer Feedback & 9-Box Talent Review Grid

TC-360-01 Multi-Rater Peer Nomination Workflow [ ] PASS [ ] FAIL

Steps: Go to **Performance** → **360 Feedback** → Click **"Nominate Reviewers"** → Select 2 peers and 1 direct report → Submit nomination.

> **Expected Result:** Nominees receive automated feedback requests; review status updates to "*Nominations Pending*".

TC-9BX-01 9-Box Talent Matrix Visualization [ ] PASS [ ] FAIL

Steps: Open **9-Box Talent Grid** tab under Performance → View employee placement along Performance (X) and Potential (Y) axes.

> **Expected Result:** 3x3 matrix renders employee cards into proper quadrants (e.g. "*High Potential / Star*", "*Core Performer*").

Test Suite 14: Priority Announcements & Compliance Read Receipts

TC-ANN-01 Publish Critical Announcement with Mandatory Read Receipt [ ] PASS [ ] FAIL

Steps: Sign in as Admin → Navigate to **Announcements** (`/announcements`) → Click **"+ New Announcement"** → Title: "*Updated UAE National Day Holiday Schedule*", Priority: "*Critical*" → Check **"Require Mandatory Read Receipt"** → Click **Publish**.

> **Expected Result:** Announcement is posted with high-priority crimson badge and banner notification sent across portals.

TC-ANN-02 Employee Acknowledgment & Admin Audit Tracking [ ] PASS [ ] FAIL

Steps: Sign in as employee Fatima → Open announcement banner → Click **"I Acknowledge & Understand"** → Switch back to Admin announcements view.

> **Expected Result:** Employee view marks notice as acknowledged. Admin readers drawer records Fatima Al-Nuaimi's name, email, and exact timestamp in compliance audit log.

Test Suite 15: Corporate Quick Links Directory

TC-LNK-01 Admin Creates Corporate Quick Link & Category Filter [ ] PASS [ ] FAIL

Steps: Navigate to **Corporate Links** (`/company-links`) → Click **"+ Add Quick Link"** → Title: "*MOHRE Official Portal*", URL: `https://www.mohre.gov.ae`, Category: `Government Portals` → Save.

> **Expected Result:** Link card renders in Government category with favicon and direct launch button.

Test Suite 16: Community Social Feed, Direct Media Upload & Pulse Polls (Feature #64)

TC-COM-01 Direct Social Binary Image Upload with Client Thumbnail Preview [ ] PASS [ ] FAIL

Steps: Navigate to **Community & Feed** (`/community`) → In post creator, enter: "*Celebrating our Q3 Milestone!*" → Click **"Attach Image"** → Select local image file.

> **Expected Result:** Instant thumbnail preview displays before publishing. Clicking **Post** uploads binary image via `POST /api/community/upload` and renders post with full image on feed.

TC-COM-02 Interactive Employee Pulse Poll Creation & Live Voting [ ] PASS [ ] FAIL

Steps: Open **Polls** tab → Create Poll: "*Preferred Day for Monthly Townhall?*" with options: `Tuesday, Thursday` → Cast vote as employee.

> **Expected Result:** Live percentage bars update dynamically without page refresh; duplicate voting is prevented per employee.

Test Suite 17: Surveys, Research-Validated Question Packs & eNPS (Feature #53)

`TC-SRV-01` 1-Click Load Research-Validated Question Pack

`[ ] PASS` `[ ] FAIL`

Steps: Navigate to **Surveys & eNPS** (`/surveys`) → Click **"+ Create Survey"** → Click **"Load Question Pack"** → Select **Gallup Q12 Engagement Pack**.

> **Expected Result:** API endpoint `GET /api/surveys/template-questions` responds. Survey builder auto-populates all 12 standardized Likert-scale questions instantly.

`TC-SRV-02` Employee Response Submission & Admin Sentiment Analytics

`[ ] PASS` `[ ] FAIL`

Steps: Sign in as employee Fatima → Open pending survey → Answer questions and submit → Return to Admin survey dashboard.

> **Expected Result:** Response recorded anonymously. Admin survey view displays calculated eNPS score (+45), response rate percentage, and breakdown charts.

Test Suite 18: 30-Day Passport & UAE Visa Renewal Notification Engine

`TC-RNW-01` Expiry Detection & 30-Day Alert Banner

`[ ] PASS` `[ ] FAIL`

Steps: Set an employee's passport expiry to 20 days from today in employee master data → Open **Notifications & Alerts** (`/notifications`).

> **Expected Result:** Alert banner renders: "*30-Day Visa/Passport Renewal Notice*" with countdown pill. Employee portal displays corresponding banner with direct upload link.

Test Suite 19: Ad-Hoc Report Builder & Scale Operations

`TC-RPT-01` Custom Report Builder with Drag-and-Drop Columns & CSV Export

`[ ] PASS` `[ ] FAIL`

Steps: Navigate to **Reports** → **Report Builder** (`/reports/builder`) → Select entity: `Employees` → Select columns: `Full Name, Basic Salary, Visa Expiry` → Click **"Export CSV"**.

> **Expected Result:** CSV file downloads with exact selected columns and filtered dataset.

Test Suite 20: Role-Based Access Control (RBAC) Permissions Matrix

`TC-RBC-01` Granular Permission Grant & Route Guard Enforcement

`[ ] PASS` `[ ] FAIL`

Steps: Navigate to **Settings** → **Permissions** (`/settings/permissions`) → Revoke `payroll.view` from Manager role → Sign in as Manager → Attempt to access `/payroll`.

> **Expected Result:** Navigation link to Payroll is hidden from Manager sidebar; direct route access redirects safely to `/dashboard` with 403 Forbidden alert.

Test Suite 21: Full End-to-End Employee Lifecycle Flow

`TC-E2E-01` Candidate Hire → Offer → Induction → Attendance → Payroll

`[ ] PASS` `[ ] FAIL`

Steps:
1. **Recruitment:** Move candidate to "*Offered*", generate and sign offer letter.
2. **Hiring:** Convert candidate to employee → System creates record & fires welcome email.
3. **Induction:** Sign in as new hire → Complete onboarding packet wizard & digital signature.
4. **Operations:** Punch attendance clock → Submit 2-day leave request → Manager approves via multi-tier routing.
5. **Performance:** Set quarterly goal with evidence attachment → Nominate 360 peer reviewers.
6. **Surveys:** Complete Gallup Q12 pulse survey anonymously.
7. **Payroll:** Admin runs end-of-month payroll cycle → Generates WPS SIF file → Employee views Total Rewards PDF.

> **Expected Result:** Complete journey succeeds with zero data loss, zero breaking errors, and 100% database persistence across PostgreSQL tables.

3. Automated Health & Endpoint Test Script

You can execute this PowerShell script directly in your terminal to verify API health across all critical endpoints:

```
$base = "http://localhost:5000"
$login = @{ email = "admin@digitaldive.demo"; password = "demo123" } | ConvertTo-Json
$res = Invoke-RestMethod -Uri "$base/api/auth/login" -Method Post -Body $login -ContentType "application/json"
$h = @{ Authorization = "Bearer $($res.token)" }

Write-Host "[1/7] Testing Survey Template Packs..." -ForegroundColor Cyan
$srv = Invoke-RestMethod -Uri "$base/api/surveys/template-questions" -Headers $h
Write-Host " -> OK. Found $($srv.Count) research question packs." -ForegroundColor Green

Write-Host "[2/7] Testing Custom Fields Schema..." -ForegroundColor Cyan
$fields = Invoke-RestMethod -Uri "$base/api/custom-fields" -Headers $h
Write-Host " -> OK. Found $($fields.Count) custom attributes." -ForegroundColor Green

Write-Host "[3/7] Testing Approval Rules Engine..." -ForegroundColor Cyan
$rules = Invoke-RestMethod -Uri "$base/api/approvals/rules" -Headers $h
Write-Host " -> OK. Loaded $($rules.Count) routing rules." -ForegroundColor Green

Write-Host "[4/7] Testing Corporate Links..." -ForegroundColor Cyan
$links = Invoke-RestMethod -Uri "$base/api/company-links" -Headers $h
Write-Host " -> OK. Found $($links.Count) bookmark links." -ForegroundColor Green

Write-Host "[5/7] Testing Total Rewards ESS..." -ForegroundColor Cyan
$rewards = Invoke-RestMethod -Uri "$base/api/ess/1/total-rewards" -Headers $h
Write-Host " -> OK. Total Package Value: AED $($rewards.totalRewardsValue)" -ForegroundColor Green

Write-Host "[6/7] Testing Pre-WPS Compliance Audit..." -ForegroundColor Cyan
$wps = Invoke-RestMethod -Uri "$base/api/payroll/pre-wps-audit" -Headers $h
Write-Host " -> OK. Audited $($wps.totalEmployees) employee records." -ForegroundColor Green

Write-Host "[7/7] Testing Announcements & Read Receipts..." -ForegroundColor Cyan
$ann = Invoke-RestMethod -Uri "$base/api/announcements" -Headers $h
Write-Host " -> OK. Loaded $($ann.Count) announcements." -ForegroundColor Green

Write-Host "ALL 7 END-TO-END SUITES PASSED CLEANLY!" -ForegroundColor Yellow
```